

Notes Week 5

a. Varying Prescriptions of $2M_{\odot}$ Stars

a.1 Wind Prescriptions

I have already simulated a detailed TPAGB star using the [Trabucchi et al. \(2018\)](#) mass-loss prescription.

To compare how the mass-loss prescription affects the evolution of TPAGB stars, I simulate additional $2M_{\odot}$ stars with the same MESA options, except a different mass-loss prescription.

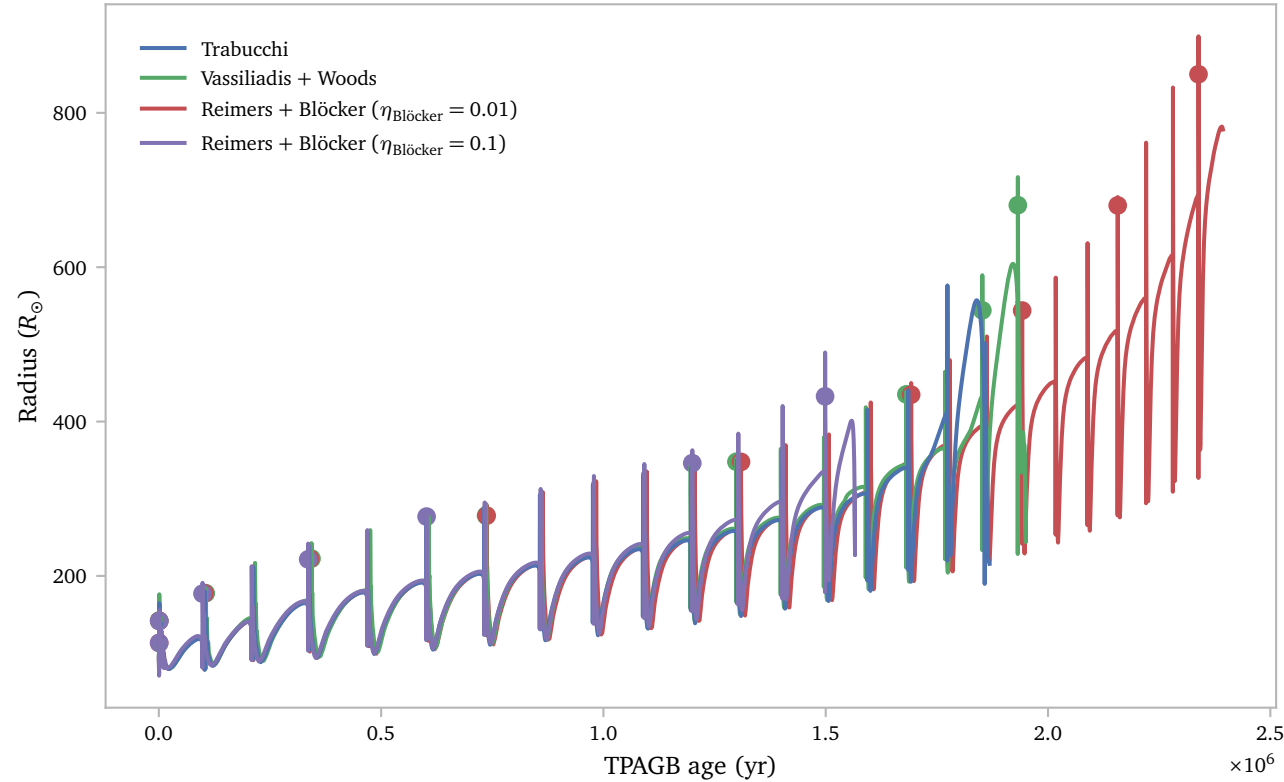
I start by simulating a star using *Reimers + Blöcker* prescriptions with the same parameters as [Cinquegrana et al. \(2022\)](#). To be explicit, they use:

```
cool_wind_RGB_scheme = 'Reimers'
cool_wind_AGB_scheme = 'Blocker'
RGB_to_AGB_wind_switch = 1.0d-4
Reimers_scaling_factor = 0.477d0
Blocker_scaling_factor = 0.01d0
```

I then simulate a star using the same Reimers + Blöcker prescription, now with the "standard" MESA Blöcker scaling factor:

Blocker_scaling_factor = 0.1d0 Lastly, I simulate a star with the Vassiliadis + Woods prescription. This is done by modifying the other_winds hook from [Rees et al. \(2024\)](#).

Below, I show the radial evolution for the 4 prescriptions.

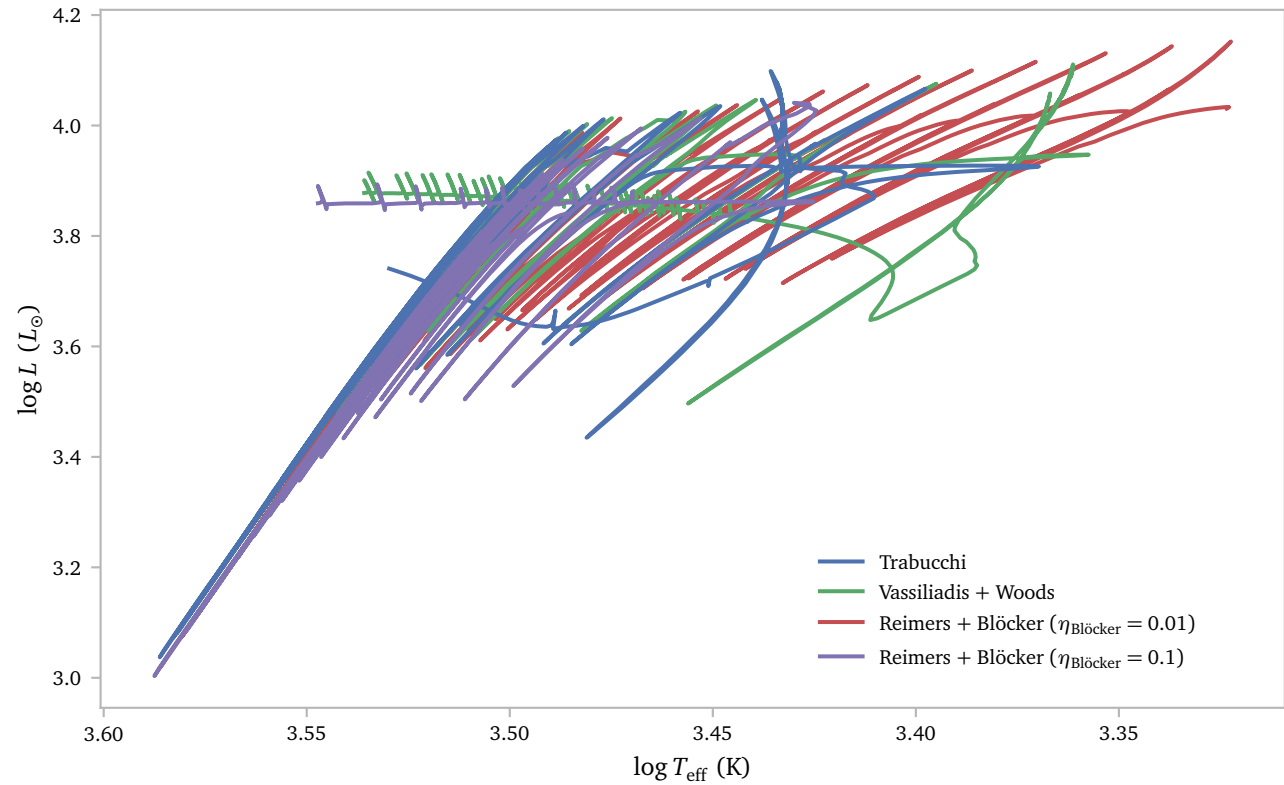


For the first few pulses, the radii of all tested mass loss prescriptions agree very well. However, once significant portions of the envelope mass is lost, the pulses will become stronger. Because the different mass loss prescriptions cause the star to (relatively) quickly lose its mass at different times in the evolution, the number of thermal pulses, and maximum radius that is achieved is very different for the different mass loss prescriptions.

The Reimers + Blöcker using the Blöcker scaling factor from [Cinquegrana et al. \(2022\)](#) terminates before expelling all of its envelope, because the AESOPUS opacity table [Cinquegrana et al. \(2022\)](#) could not find values and eventually the time step dropped below the minimum timestep.

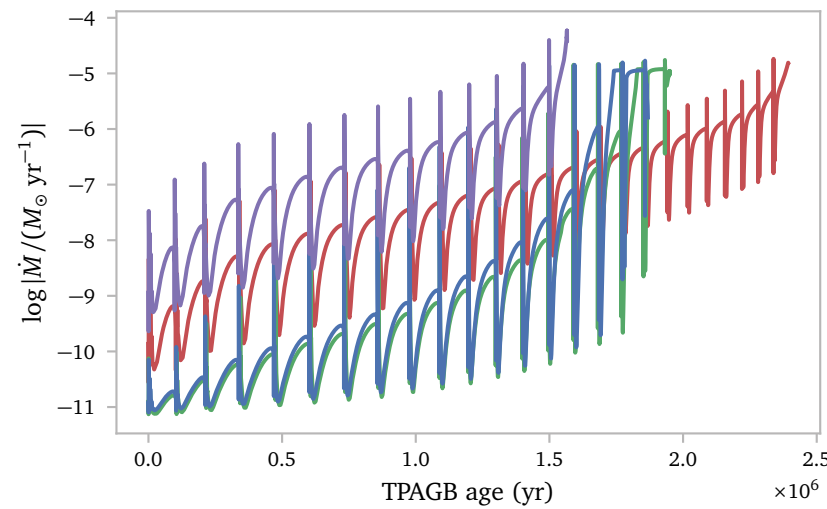
The other models terminate after exceeding the minimum envelope mass.

Below I plot the HR diagram of the four models. The plot is very messy, but I think two things are interesting.



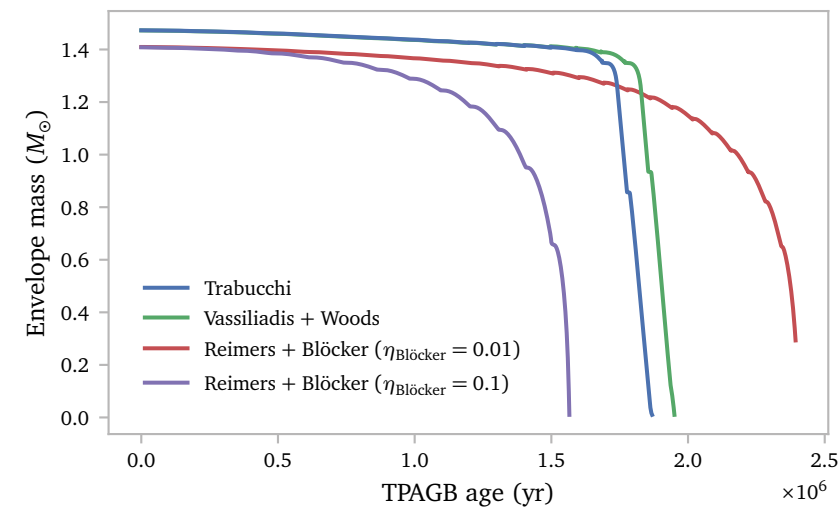
- The last few pulses of the AGB star cause it move (to me) in unpredictable ways.
- The mass-loss prescription from Trabucchi leads to a different approach to the post-AGB phase in the HR diagram.

Below, I plot the absolute mass loss rate of the star for the various mass-loss prescriptions.



The colors are the same as in the figure above. The two models using Reimers + Blöcker have the same general shape. The other two models also have the same general shape.

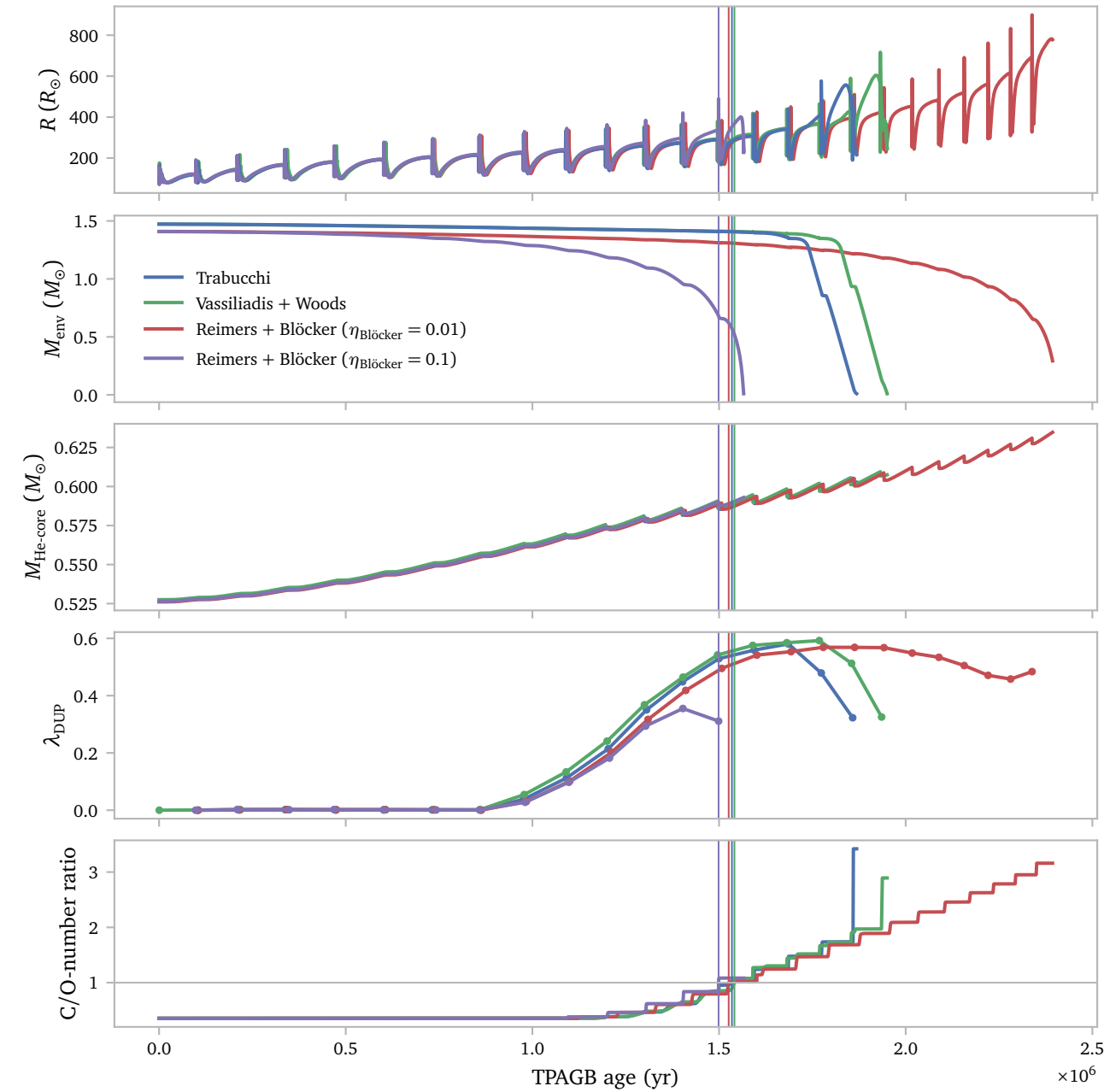
Below, I plot the envelope mass of the star for the various mass-loss prescriptions.



Again, the differences between the mass loss prescriptions are obvious.

The models using Reimers + Blöcker start at a lower mass, because the mass loss rate is higher during the previous giant evolutionary phases.

Lastly, I compare the dredge up of the different models.



Here, a couple of things are interesting.

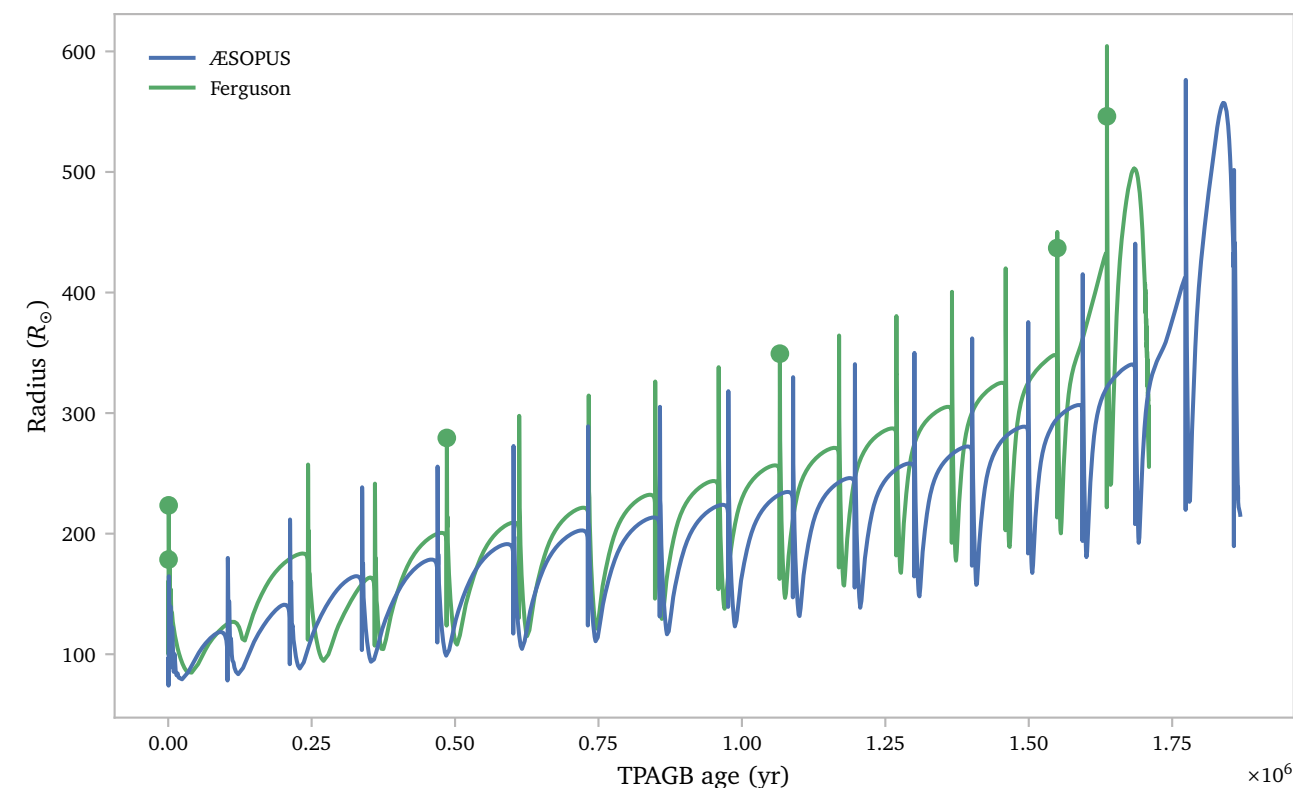
- The helium core mass is nearly identical for all models.
- The dredge-up efficiency λ_{DUP} seems to be more efficient when a larger part of the envelope is intact, meaning initially, the models using Reimers + Blöcker mass loss prescription have slightly lower efficiencies. After the fast mass loss phase starts, the efficiency quickly drops below that of the Cinquegrana et al. (2022) Reimers + Blöcker mass loss prescription.
- The C/O-ratio makes significantly larger jumps when using the Trabucchi or Vassiliadis + Woods prescription.

I am not sure which mass-loss prescription is *best*. I opt to use the Vassiliadis & Wood (1993) prescription for two reason.

1. For a $2M_{\odot}$ star, this prescription seems to be a middle ground between the prescription used by Rees et al. (2024) and the prescription used by Cinquegrana et al. (2022).
2. I already have models for this star, while I do not have models for the star using the Trabucchi et al. (2018) prescription.

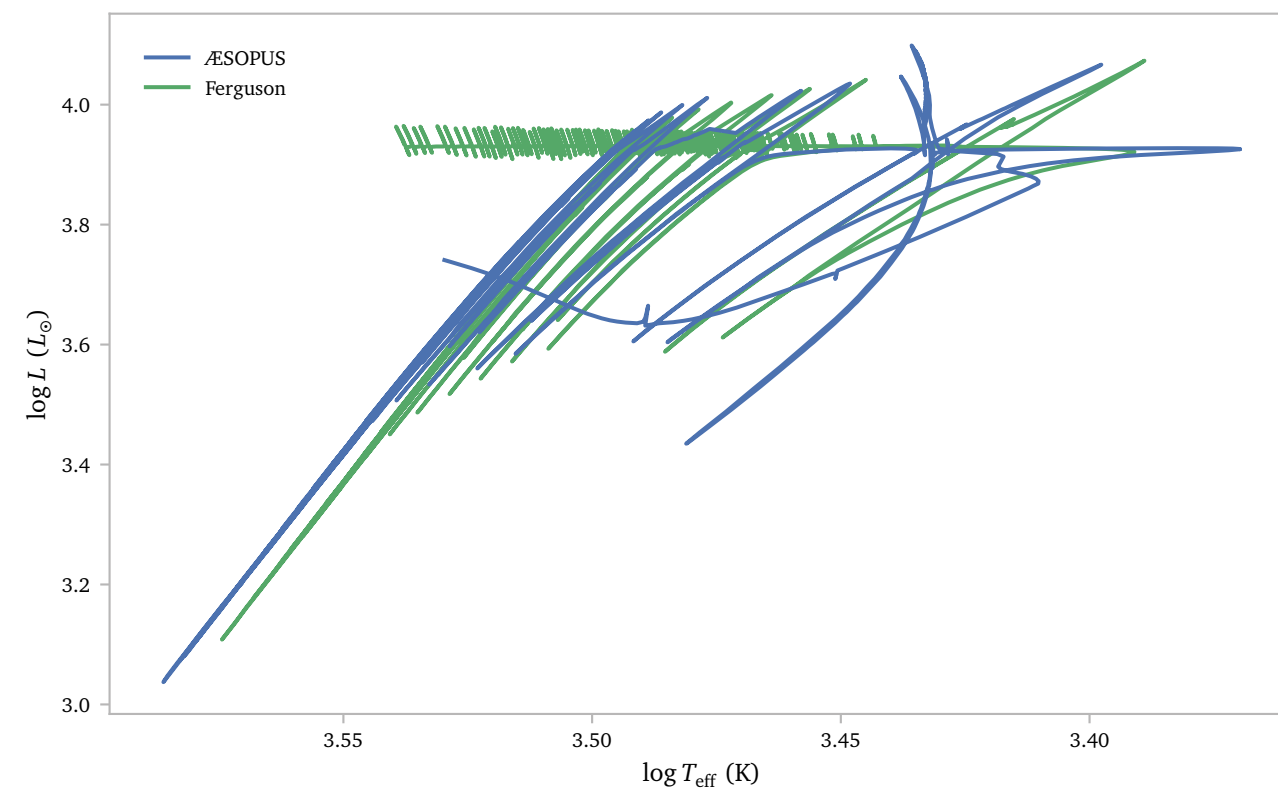
a.2 Opacity Tables

Because some models seem to have trouble finding opacities in the *ÆSOPUS* opacity table, I thought it would be interesting to compare how much difference using a different table would make. I compare *ÆSOPUS* from [Marigo & Aringer \(2009\)](#) with the [Ferguson et al. \(2005\)](#) opacities for a [Asplund et al. \(2009\)](#) solar model, which is assumed in the high T regime as well.



Clearly, but not surprisingly, using a different opacity table has profound effects on the radius of the star. Apparently, the opacities of [Ferguson et al. \(2005\)](#) are slightly higher, resulting in a larger radius, especially for a very large convective envelope.

Below I show the HR diagram of the two models.



Unlike the different approach to the post-AGB phase that the *ÆSOPUS* model has for the [Trabucchi et al. \(2018\)](#) mass-loss prescription, this approach is not present when using the [Ferguson et al. \(2005\)](#) opacities.

Since *ÆSOPUS* from [Marigo & Aringer \(2009\)](#) specifically takes into account the varying C/O ratio, and because this table is used in [Rees et al. \(2024\)](#), I decide to try mass transfer using this opacity table, *but* this might mean I encounter difficulties with out of bound parameters.

b. Quasi-Adiabatic limit

b.1 Implementation

I need to implement the quasi-adiabatic stopping condition from Temmink et al. (2023) in the run_binary_extras code¹. This condition gives an upper limit for the mass transfer rate. I want to terminate the binary models when the mass transfer rate exceeds the condition. To achieve this, I implement the following code in the integer function extras_binary_finish_step(binary_id).

For the quasi-adiabatic stopping condition from Temmink et al. (2023), we need the local thermal timescale. MESA internally saves this value in profile data when this is enabled. I found the calculation of the local thermal timescale in the file profile_getval.f90:

```
! profile_getval.f90
subroutine getval_for_profile(s, c, k, val, int_flag, int_val)
...
  case (p_thermal_time_to_surface)
    if (s% L(1) > 0) &
      val = sum(s% dm(1:k)*s% cp(1:k)*s% T(1:k))/s% L(1)
...
end subroutine getval_for_profile
```

I implement this same definition. It is the discrete version of the thermal timescale definition given in Temmink et al. (2023). The local thermal time is then used to calculate the critical mass loss rate using equation 8 of Temmink et al. (2023). Below I show the code I added in integer function extras_binary_finish_step (binary_id)

```
integer function extras_binary_finish_step(binary_id)
  type (binary_info), pointer :: b
  type (star_info), pointer :: s1
  integer, intent(in) :: binary_id
  integer :: ierr

  logical :: do_termination

  real(dp) :: cum_energy, tau_local, Mdot_crit, max_Mdot_crit
  integer :: k

  call binary_ptr(binary_id, b, ierr)
  call star_ptr(b% star_ids(1), s1, ierr)
  if (ierr /= 0) then ! failure in binary_ptr
    return
  end if

  extras_binary_finish_step = keep_going

  do_termination = .false.

  cum_energy = 0d0
  max_Mdot_crit = -1d99

  do k = 1, s1% nz
    cum_energy = cum_energy + s1% dm(k) * s1% cp(k) * s1% T(k)

    if (s1% L(1) <= 0d0) cycle
    tau_local = cum_energy / s1% L(1)

    if (tau_local <= 0d0) cycle
    Mdot_crit = (s1% mstar - s1% m(k)) / Msun / (tau_local / secyer)
    max_Mdot_crit = max(max_Mdot_crit, Mdot_crit)
  end do

  write(*,'(A,1PE12.5)') 'critical Mdot = ', max_Mdot_crit
  write(*,'(A,1PE12.5)') 'current Mdot = ', -s1% star_mdots

  if (-s1% star_mdots > max_Mdot_crit ) then
    do_termination = .True.
  end if

  if (do_termination .eqv. .True.) then
    termination_code_str(t_xtral) = "Critical mass loss rate exceeded"
    s1% termination_code = t_xtral
    extras_binary_finish_step = terminate
  end if

end function extras_binary_finish_step
```

b.2 A simple test

As a test, I take the EAGB mass transfer model that I ran last week, and I modify the quasi-adiabatic condition such that it triggers much earlier.

The EAGB model nicely terminates once the mass loss rate exceeds the critical mass loss rate:

```
critical Mdot = 8.40666E-12
current Mdot = 9.15848E-12
save LOGS/EAGB/profile1.data for model 99

max increase dt limit 100

bin ...
save photos/b_x099, photos/EAGB/1_x099 for model 99

runtime (minutes), retries, steps 1.71 0 99
termination code: Critical mass loss rate exceeded

-----
step ...
-----
99 ...
min_beta 0.8937377141189621
alpha_mlt_max 1.9310000000000000
alpha_mlt_surface 1.9310000000000000

DATE: 2026-03-01
TIME: 01:02:26
finished
```

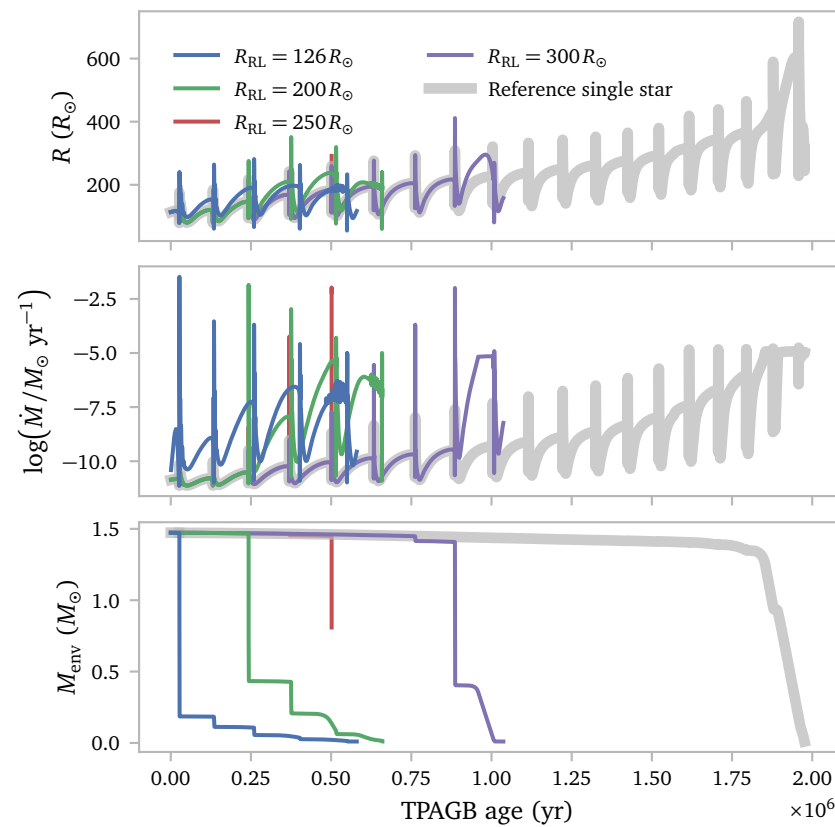
¹ This is the binary equivalent for the run_star_extras, where extra methods can be implemented before, during and after binary steps, and the binary pointer b% containing the binary structure.

c. Testing TPAGB stars

I ran 15 models of TPAGB stars. I chose 4 different (initial) RLOF radii and simulated 4 different q values for each radius². The smallest RLOF radius was chosen such that the initial radius of the starting model is $0.9 \times$ the radius of the initial Roche lobe. The starting model has a radius of $R \approx 113R_{\odot}$ and as such, the initial RLOF radius of the model is $R_{\text{RL}} \approx 126R_{\odot}$. For this radius, I simulated $q = 1, 0.75$ and 0.50 .

Then, I simulated binary models for three additional initial RLOF radii: $R_{\text{RL}} \approx 200, 250$ and 300 . For all of these, I simulated $q = 1.0, 0.75, 0.50$ and 0.25 .

Below, I show some properties of the binary evolution for $q = 1.0$.



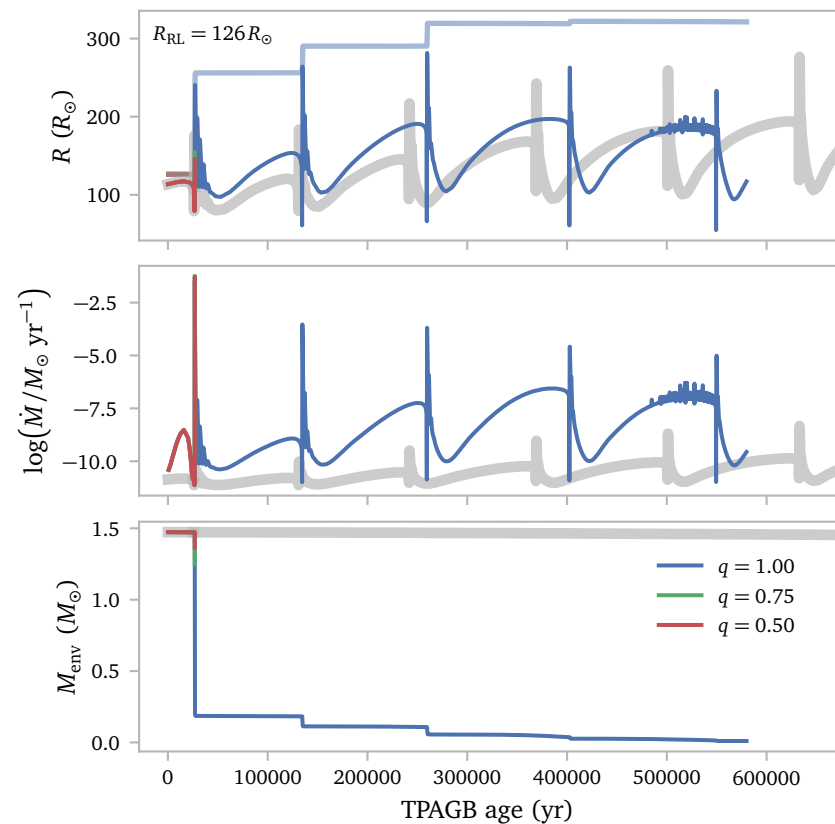
² Except for the smallest RLOF radius. This model had stability problems. I will come back to this.

The $q = 1$ model for $R_{\text{RL}} = 250R_{\odot}$ terminated early due to exceeding the RLOF retry limit.

The first pulse where $R \approx R_{\text{RL}}$ generally seems to result in the highest mass loss rate and the most amount of mass lost. The pulses following the strong pulse are less extreme and transfer less mass.

For some models, especially more evolved TPAGB models, the stars lose significant amounts of mass between pulses due to winds.

Now I show the binary models with $R_{\text{RL}} = 126R_{\odot}$

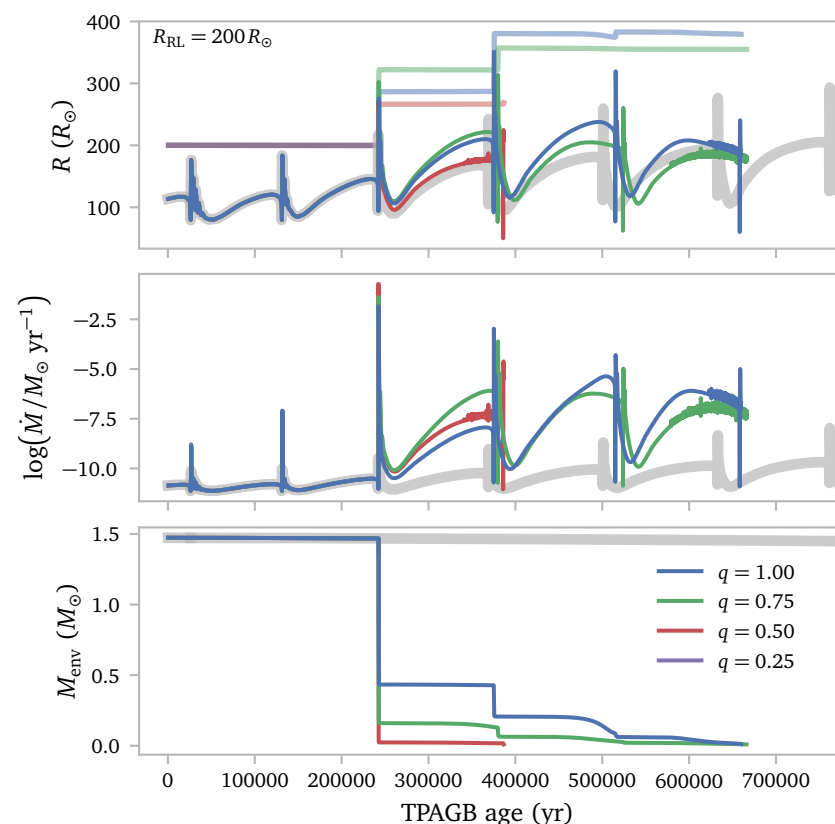


The transparent line shows the evolution of the Roche lobe radius of the donor.

The first pulse transfers a significant amount of mass. The following pulses only a very small amount. The Roche Lobe radius jumps after every pulse. Eventually, the model terminates due to *wind* mass loss.

The $q = 1$ model terminates due to exceeding the minimum envelope mass limit. This is great. The other two models however, terminate due to repeated problems finding suitable input values for the opacity table. After many retries, the minimum Δt is exceeded and the models terminate.

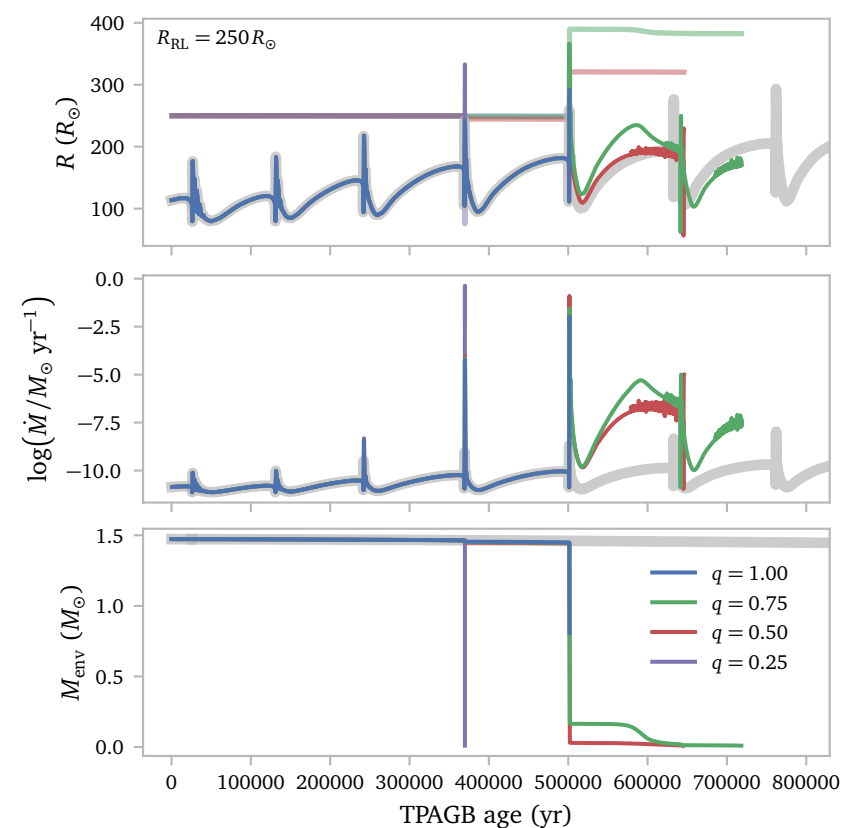
Below is the same plot, now for the $R_{\text{RL}} = 200R_{\odot}$ models.



The $q = 0.25$ model terminates due to repeated problems finding suitable input values for the opacity table. Again, after many tries, the minimum Δt exceeds the minimum value set. This seems to be the same problem that caused the models for the $R_{\text{RL}} = 126R_{\odot}$ case to terminate.

The low q models have stronger mass transfer. They also transfer more mass during the first substantial pulse.

Below is the same plot, now for the $R_{\text{RL}} = 250R_{\odot}$ models.

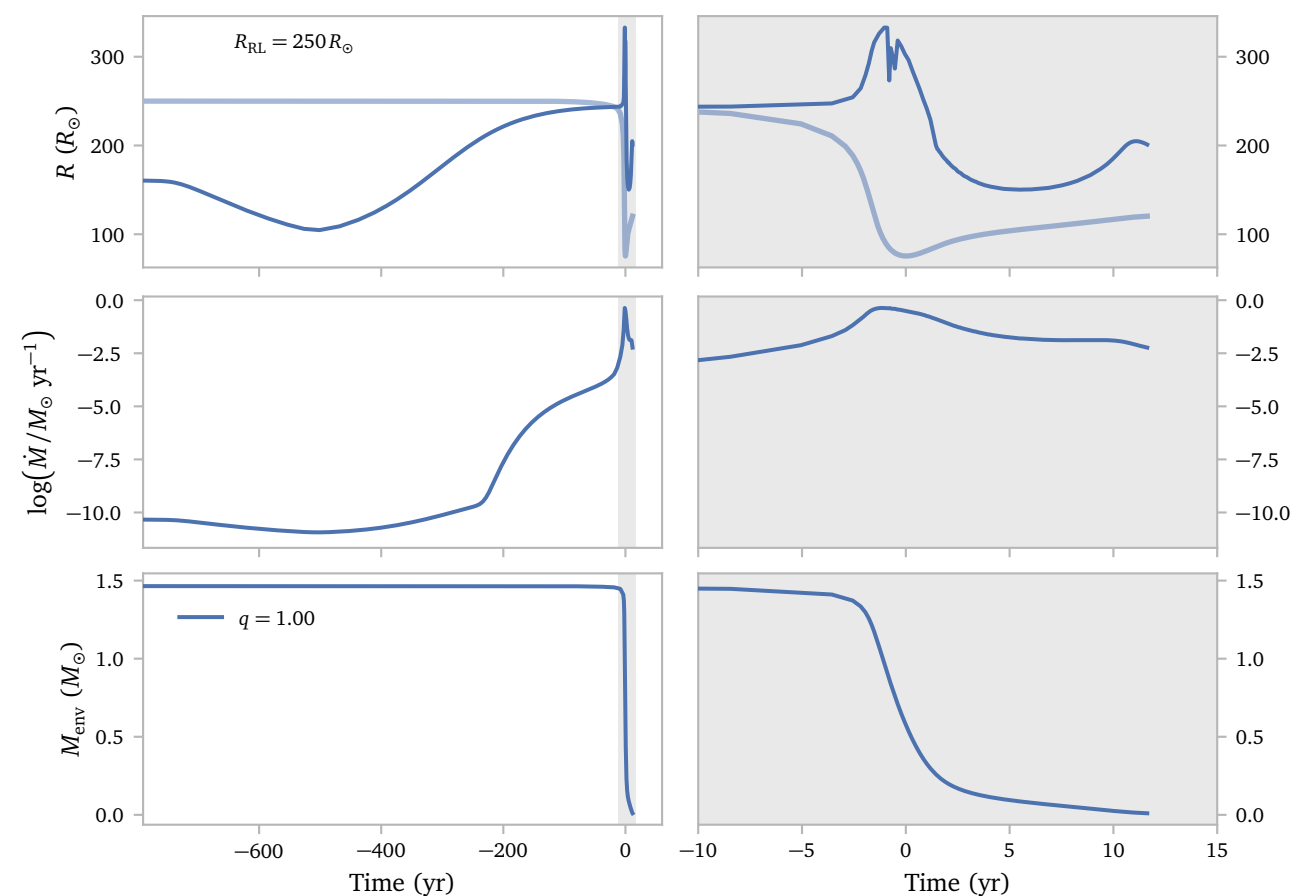


This time, the $q = 1$ model failed to terminate due to exceeding the envelope mass limit. The solver gave up after requiring more than 100 iterations to find the mass transfer rate due to RLOF. The other models ran successfully.

I think it is interesting to see that, depending on the mass ratio q , significant amounts of mass transfer will occur during different pulses.

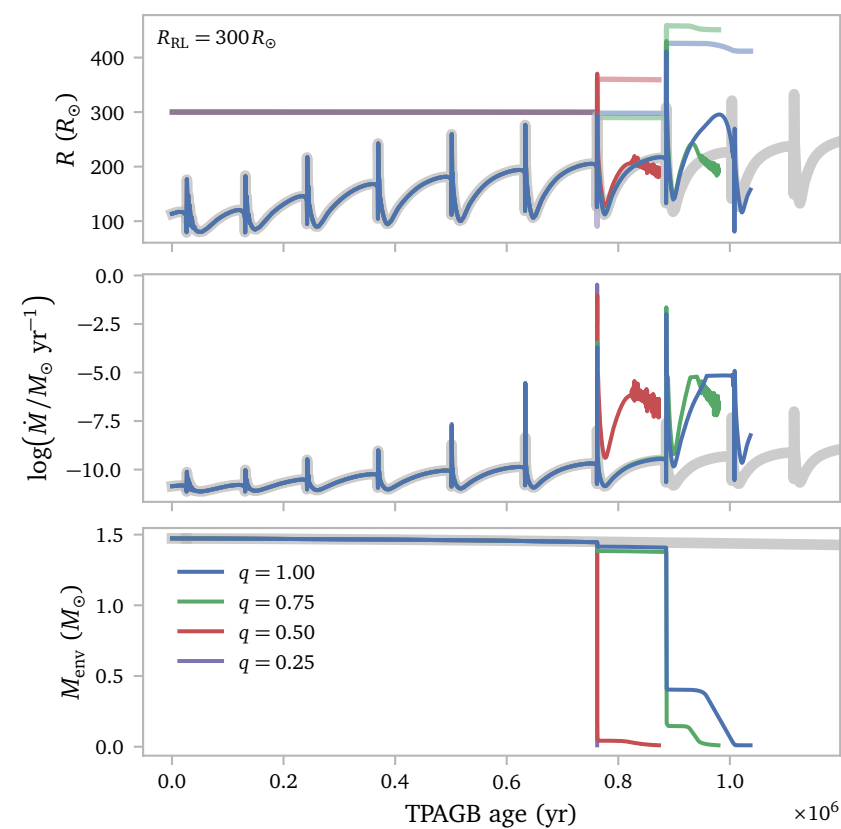
For the $q = 0.25$ model, the mass transfer rate is so strong during the first significant pulse, that it sheds its entire envelope during this single pulse. However, the mass loss rate *never* exceeds the critical mass loss rate. The maximum mass loss rate was about $0.43 M_{\odot} / \text{yr}$ while the critical mass loss rate at that point is $2.5 M_{\odot} / \text{yr}$.

The plot below zooms in on the pulse that sheds the entire envelope at once.



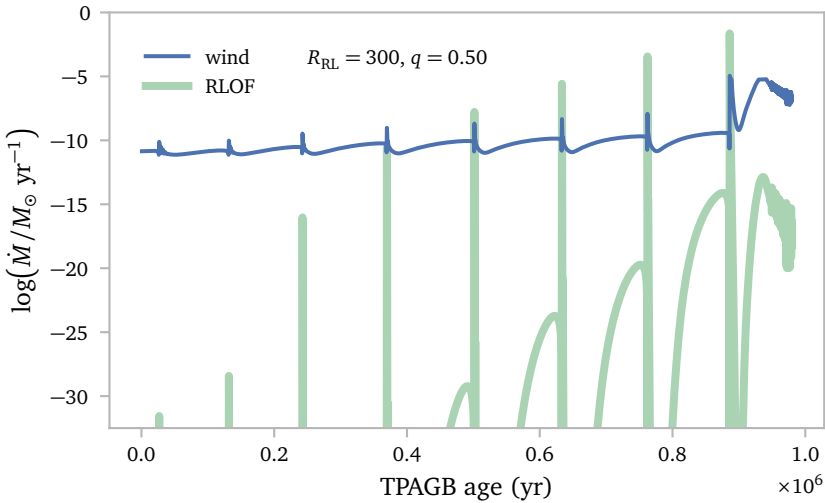
To me, it seems concerning that at one point, the radius is more than $3 \times$ bigger than the Roche lobe at the most extreme point.

Below, I plot the binary evolution for the models with an initial Roche lobe radius of $R_{\text{RL}} = 300R_{\odot}$.



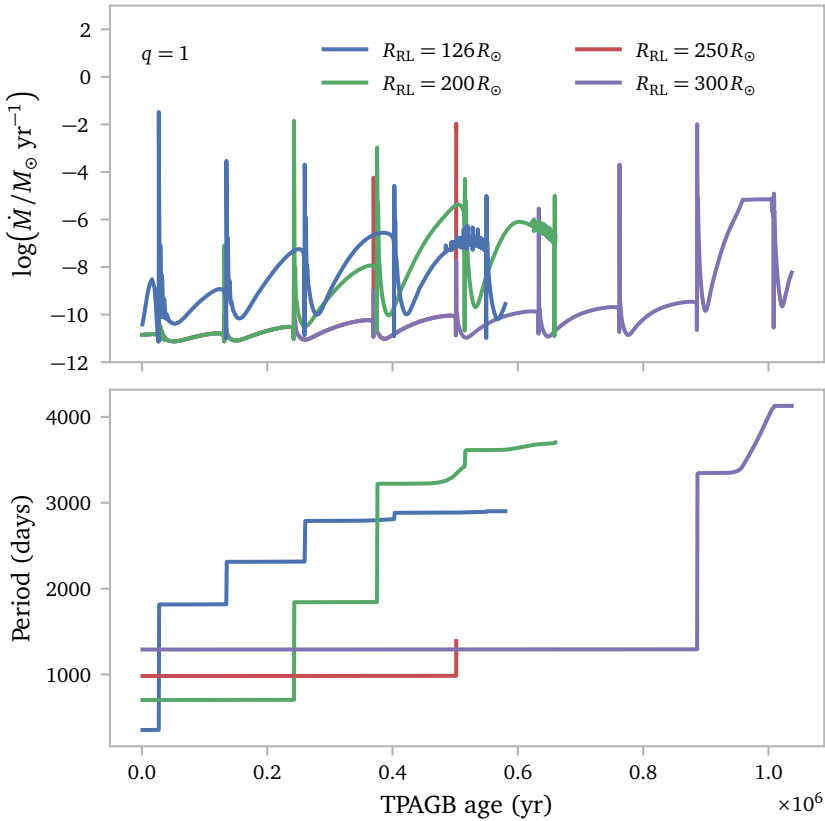
All models terminate due to exceeding the minimum envelope mass. The models with $q = 0.25$ and 0.50 undergo their first phase of significant mass transfer one pulse earlier than the other two models.

To see what part of the mass loss is due to winds, and what part is due to RLOF mass transfer, I plot the respective parts below.



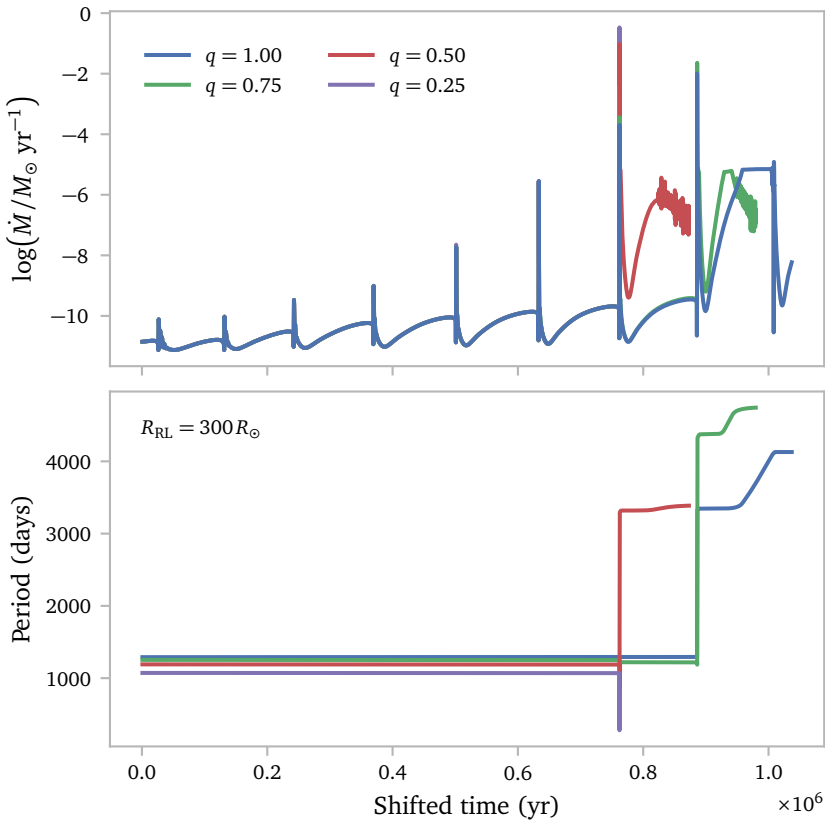
During the interpulse period, mass loss is dominated by winds. During (later) pulses, mass loss is dominated by RLOF mass transfer. This aligns well with the plots showing the mass of the envelope, where significant drops in envelope mass can be seen during some of the pulses.

Below, I plot the evolution of the orbital period of the binary for the $q = 1$ systems.



The orbital period drastically increases due to the mass transfer and the jumps align with the periods of fast RLOF mass transfer.

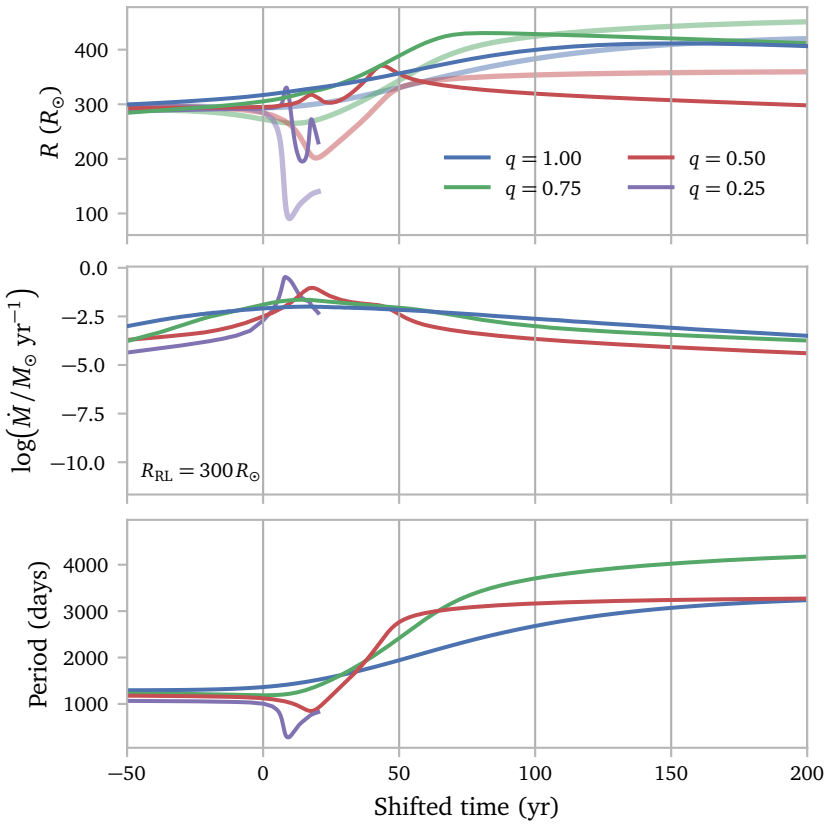
There are some important slight differences for different mass ratios though. Below I plot:



From this sample, it looks like systems with lower q end up as shorter period binaries.

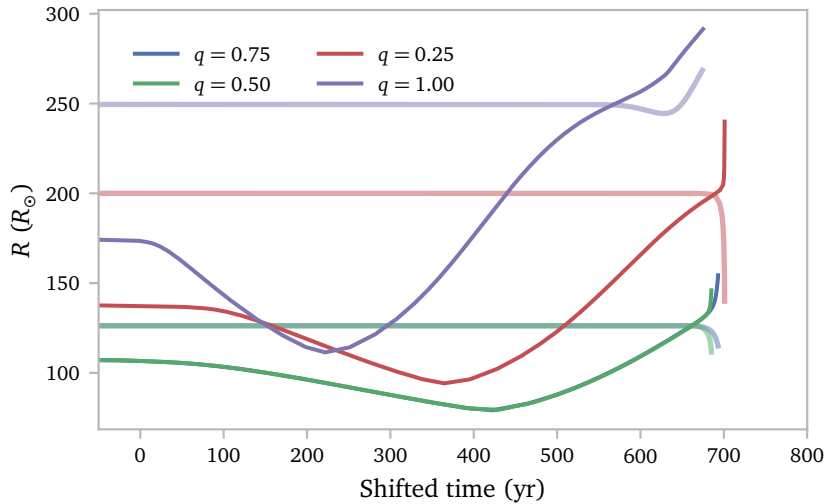
Systems with q significantly lower than 1 also tend to first undergo a phase in which the orbit shrinks, after expanding again.

Below I plot a zoom in on the strongest pulse for these systems.



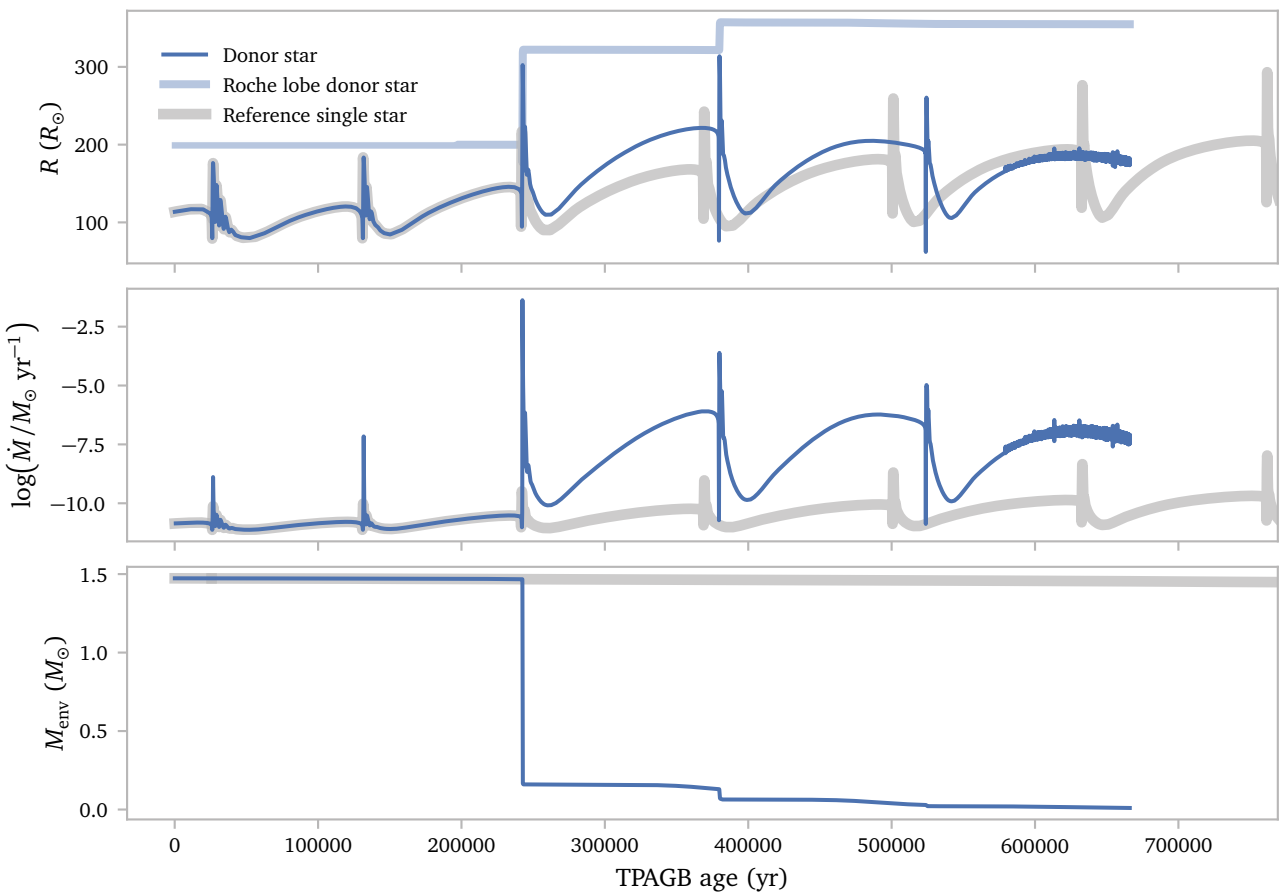
I also add the radius and Roche lobe radius. Here, the drop in period is much more visible. It aligns very well with the drop in Roche Lobe radius while the stellar radius tends to increase at this point. This is also the point with the strongest mass loss.

Now. I plot the radius of the failed models:



Apart from the $q = 1$ model that failed due to the RLOF iteration limit, the other models all fail when the Roche lobe radius is dropping.

Below, I plot the side deck figure for completeness.



d. *ÆSOPUS*

When the following message is printed to the terminal,
AESOPUS_interp: logR, logT outside of table bounds,
MESA has internally called this function and evaluated it to .false.:

```
! kap/private/kap_aesopus.f90
logical function outside_R_and_T_bounds(logR,logT)
  real(dp), intent(in) :: logR, logT
  outside_R_and_T_bounds = &
    logR < kA% min_logR .or. logR > kA% max_logR .or. &
    logT < kA% min_logT .or. logT > kA% max_logT
end function outside_R_and_T_bounds
```

Here, $R = \rho / (T / 10^6 \text{ K})^3$.
Unfortunately, I do not have access to the surface density, so I cannot easily
compute R for models that fail.
I am going to include this in the next models I run.

References

Asplund, M., Grevesse, N., Sauval, A. J., & Scott, P. (2009) *The Chemical Composition of the Sun*. [ARA&A47, 481](#).

Cinquegrana, G. C., Joyce, M., & Karakas, A. I. (2022) *Bridging the Gap between Intermediate and Massive Stars. I. Validation of MESA against the State-of-the-Art Monash Stellar Evolution Program for a 2M_⊙ AGB Star*. [ApJ939, 50](#).

Ferguson, J. W., Alexander, D. R., Allard, F., et al. (2005) *Low-Temperature Opacities*. [ApJ623, 585](#).

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Rees, N. R., Izzard, R. G., & Karakas, A. I. (2024) *Thermal pulses with MESA: resolving the third dredge-up*. [MNRAS527, 9643](#).

Temmink, K. D., Pols, O. R., Justham, S., Istrate, A. G., & Toonen, S. (2023) *Coping with loss. Stability of mass transfer from post-main-sequence donor stars*. [A&A669, A45](#).

Trabucchi, M., Wood, P. R., Montalbán, J., et al. (2018) *Long-Period Variables in the Large Magellanic Cloud*. [514, 111](#).

Vassiliadis, E. & Wood, P. R. (1993) *Evolution of Low- and Intermediate-Mass Stars to the End of the Asymptotic Giant Branch with Mass Loss*. [ApJ413, 641](#).